

HD in Computing & Software Engineering

CSE5015 Computing Project Proposal Approval Sheet

Student Use only;

Group Name	Group 07	Attempt No	1
Title of the project	CreditFlow- Automated Loan Restructuring System		
System platform	☒ Web Application ☐ Desktop Application ☐ Mobile Application		
Technology	FrontEnd-HTML, CSS, JAVASCRIPT, React.js (for dynamic UI) BackEnd-Java (Spring Boot FrameWork) DataBase-MySQL Security-Role-Based Access Control (RBAC), JWT Authentication / Session-based login Integration-REST APIs (for Core Banking System connection) AI Module-Python (for document verification)		
Programming methodology	☐ Structured System Analysis and Design Methodology ☒ Object Oriented Analysis and Design Methodology		
SDLC	Agile Software Development Life Cycle (Scrum-based iterative development)		

Final Project Proposal

Project Title

CreditFlow - Automated Loan Restructuring System

(Housing and Personal Loans – Banking Sector)

Project Background

At present, Our Bank manages loan restructuring and rescheduling requests for housing and personal loans through a largely manual and fragmented process handled primarily by the Remedial Unit, which is responsible for customers experiencing repayment difficulties.

Customers submit written request letters, after which bank officers manually recalculate repayment plans using spreadsheets. Approval memorandums are prepared manually and circulated through email, while post-restructuring monitoring is handled separately using basic reporting mechanisms.

The absence of a centralized system to manage restructuring activities end-to-end results in several operational challenges, including processing delays, calculation inconsistencies, documentation risks, limited audit traceability, and weak monitoring controls. These inefficiencies increase operational workload and create compliance risks within a highly regulated banking environment.

According to Central Bank of Sri Lanka (CBSL) regulatory guidance:

- Restructure applies to accounts below 90 Days Past Due (DPD)
- Reschedule applies to accounts above 90 Days Past Due (DPD)

Currently, classification and monitoring are handled manually by officers, increasing the possibility of inconsistency.

This project proposes the design of a centralized internal system that digitizes and automates the complete loan restructuring lifecycle while ensuring alignment with banking policies and regulatory requirements.

Project Objective

The objective of this project is to design an internal banking system that automates the restructuring and rescheduling process for housing and personal loans by introducing:

- Structured digital workflows
- Automated repayment recalculations
- AI-assisted document verification integrated with approval memo
- Electronic approval routing
- Controlled execution through credit units
- CBSL-compliant monitoring automation
- Continuous monitoring and automated customer communication

The proposed solution aims to improve operational efficiency, accuracy, compliance, and transparency within the restructuring process managed by the Remedial Unit.

Project Scope Overview

This project focuses strictly on system design and project planning. The scope includes business process modelling, functional module design, workflow definition, data handling concepts, and technology recommendations.

The project does not involve system development, deployment, or integration with live banking systems.

Proposed End-to-End Restructuring Workflow

The system is designed to manage the complete restructuring lifecycle through structured stages led by the Remedial Unit.

1. Customer Request Initiation

Any customer holding an active housing or personal loan, whether performing or delinquent, may request loan restructuring.

The restructuring process begins only after submission of a formal written request letter.

Requests may be:

- Submitted directly by customers to the officers

- Collected by bank officers during follow-ups

All request letters must be uploaded to the system. Processing cannot proceed without a verified request letter, ensuring audit traceability and regulatory compliance.

Example:

A customer currently paying an EMI of LKR 72,000 requests a reduced installment through tenure extension due to affordability concerns.

2. Restructuring Case Creation and Validation

Once the request letter is uploaded;

A restructuring case is automatically generated.

The system retrieves:

- Loan details

- Repayment history

- Arrears status

The system automatically calculates Days Past Due (DPD) and tags the case as:

- Restructure – Below 90 DPD

- Reschedule – Above 90 DPD

This automatic tagging ensures regulatory consistency and removes manual classification errors.

3. Automated Restructuring Analysis

Bank officers enter proposed restructuring parameters such as revised installment or tenure.

The system automatically calculates:

- Revised EMI

- Extended loan tenure

- Updated repayment schedule

Financial impact summary

All calculations follow predefined banking rules and internal credit policies, eliminating spreadsheet dependency and reducing calculation errors.

4. Auto-Generated Credit Approval Memo

The system automatically generates a structured approval memorandum including:

- Customer repayment behavior
- Reason for restructuring
- Existing vs proposed repayment structure
- Financial impact analysis
- Officer recommendation

The approval memo becomes the central document package for credit evaluation.

5. AI-Based Document Verification (Before Approval)

Supporting documents are submitted together with the generated approval memo, including:

- Salary slips
- CRIB reports
- Employer confirmations
- Business documents (where applicable)

An AI-assisted verification module performs:

- Document completeness checks
- Format validation
- Data consistency verification against loan records

All verification occurs before the approval stage.

Only verified documents and validated financial figures are forwarded for management approval.

Approval is granted only after successful verification of both the memo and supporting documentation.

6. Sales Unit Documentation and Customer Consent

After approval, the case is forwarded to the Sales Unit

The sales unit is responsible for:

- Explaining revised loan terms to the customer

- Completing required documentation

- Obtaining customer consent and signatures

Two formal applications are completed:

- Restructure Application

- Loan Amendment Application

Signed documents are uploaded back into the system as legal confirmation of customer agreement.

7. Credit Department Processing

Once signed documentation is uploaded:

- The case is routed to the Credit Department.’

- Credit officers finalize restructuring parameters.

- A formal restructuring advice is prepared.

System instructions are then generated for execution by the Centralized Credit Processing Unit (CCPU).

8. Restructure Execution by CCPU

The CCPU performs:

- Loan rescheduling within the core banking system

- Updating repayment terms

- Activation of revised installment structures

The restructuring becomes effective only after successful execution.

9. Post-Activation Monitoring and Communication

After activation, monitoring begins automatically according to CBSL guidelines:

Case Type	Monitoring Period
------------------	--------------------------

Restructure	90 Days
-------------	---------

Reschedule	120 Days
------------	----------

The system will:

- Provide dashboards for remedial monitoring

- Maintain regulatory compliance tracking

Automated communications include:

- Revised repayment schedule emails

Proposed Technologies (Conceptual Design)

- Web-based internal banking application

- Centralized relational database

- Rule-based restructuring calculation engine

- AI-assisted document verification module

- Role-based access control

- Automated notification services

Technology selection remains conceptual and design-focused.

System Functionality:

Function	Description
1. User Registration & Login	1. Super Admin • Manage user accounts • Assign roles and permissions • Monitor system activity • Maintain system security • Configure system settings 2. Remedial Unit User • Create restructuring cases • Upload request letters • Perform restructuring analysis • Generate approval memos • Monitor post-restructure accounts 3. Sales Unit User • Explain revised loan terms • Collect customer signatures • Upload consent documents • Confirm customer agreement 4. Credit Department User • Review restructuring proposals • Finalize loan parameters • Approve restructuring decisions • Generate restructuring advice 5. CCPU User • Execute restructuring in CBS • Update loan repayment schedules • Activate revised loan terms 6. Risk & Compliance User • Monitor compliance with CBSL rules • Review audit logs • Track monitoring periods (90/120 days) • Generate compliance reports
	• Upload customer request letter • Validate mandatory request document • Store request details in system

2.Request Submission Function	
3.Case Creation Function	• Automatically generate case ID • Retrieve loan and customer data • Initialize restructuring workflow
4. DPD Classification Function	• Calculate Days Past Due (DPD) • Auto classify: • Restructure (<90 days) • Reschedule (>90 days)
5. Financial Calculation Engine	• Calculate revised EMI • Generate repayment schedule • Adjust interest & tenure • Provide financial summary
6. Approval Memo Generation	• Auto-generate structured memo • Include: ○ customer profile ○ repayment history ○ proposed changes • Export as PDF
7. Document Management System	• Upload supporting documents • Categorize documents • Secure storage & retrieval
8. AI Document Verification	• Check document completeness • Validate format and accuracy • Compare with system data
9. Workflow & Approval Routing	• Route cases between departments • Track approval status • Escalate pending cases

10. Customer Consent Management	• Capture signed documents • Upload consent forms • Confirm customer agreement
11. Credit Processing Function	• Finalize restructuring parameters • Generate restructuring advice • Send instructions to CCPU
12. Core Banking Integration	• Send approved data to CBS • Update loan account details • Activate new repayment plan
13. Monitoring System	• Track repayment behavior • Monitor 90/120-day periods • Identify missed payments
14. Notification System	• Send EMI reminders • Notify delays and arrears • Generate alerts to officers
15. Reporting & Audit Trail	• Track all system activities • Maintain audit logs: • who did what • when changes occurred • Generate compliance reports
16. Dashboard & Analytics	• Show case status • Monitoring summaries • Performance insights